Semarchy

Modeling

Entities and attributes

ER Logical Modeling

- Entity-Relationship Modeling

  Entities are the equivalent of grammatical nouns, such as employees, departments, products, or financial accounts. An **entity** can be defined by means of its properties, called **attributes**. **Relationships** are the equivalent of verbs or associations, such as the act of purchasing, the act of repairing, being a member of a group, or being a supervisor of a department.

- **Entities**
  - are business concepts, and **<u>not</u>** tables.
  - are made of **attributes**
  - are related using **reference relationships**

Entities

- Composed of **attributes**
  - Either **simple** (built-in, LOV, user-defined type) or **complex** (complex type)
  - holds data quality rules (mandatory, LOV value range, etc.)
- **Entity type** defines how the entity matches records
  - **Fuzzy matched**: matches records from multiple sources based on data similarity.
  - **ID matched**: matches record from multiple sources using a common ID.
  - **Basic**: no matching, all records come from a single source or are authored in the data hub.
- The **primary key**
  - The attribute that identifies each **golden record**
  - Can be handled by the platform (UUID, Sequence, calculated using SemQL)
  - … or loaded from the source systems
- **Reference relationships** relate entities

Entity Type

	Basic	ID-Matched	Fuzzy-matched
Publishers	One or none	Multiple	Multiple
Match and Merge	**Not supported** Dups detected at creation time	**Yes**, using exact match on a shared ID Dups detected at creation time	**Yes**, using match rules Dups detected at creation time
Survivorship	**Not supported**	Consolidation is supported Value override is supported	Consolidation is supported Value override is supported
ID Management	Golden ID=Source ID	All source systems use a shared ID Golden ID=Source ID	Source systems have different IDs The Golden ID is generated
Recommended for	Centralized pattern Single source system Reference Data	Consolidation pattern, for entities with a shared ID.	Consolidation pattern, for entities with no shared ID.
Example	Product Family, Cost Centers	Products (with a shared SKU)	Customers, Contacts, Products (with no shared SKU)

Primary key attribute

- Primary Key (ID) identifies the master and golden records
- Behavior differs for the entity type
- **Basic and ID-Matched** entities
  - Golden and Master ID are the same
    - Generated by xDM (Sequence, UUID, Computed from other attributes)
    - or Manually entered/provided by source systems
- **Fuzzy-Matched** entities
  - Master and Golden records have a different IDs.
  - Golden ID
    - Generated by xDM (Sequence or UUID)
  - Master ID
    - Generated by xDM (Sequence, UUID, Computed from other attributes)
    - or Manually entered/provided by source systems

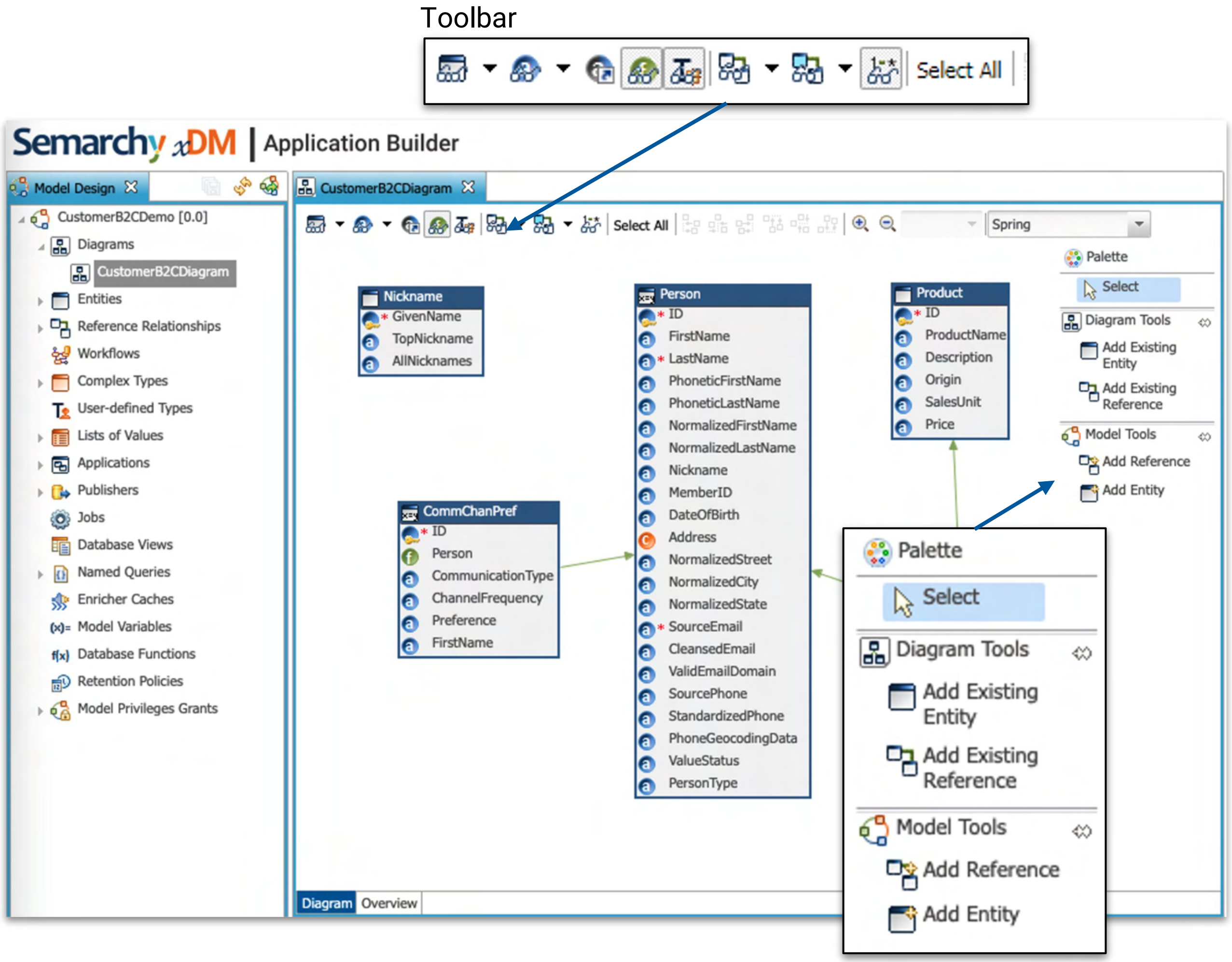
Names and Labels in Entities and Attributes

- Define simple and explicit **names** for entities and attributes
  - The name is used in SemQL rules and in the REST APIs.
  - E.g.: Customer, Address, BankDetail, etc.
- Define good **labels** and **plural labels** for entities and attributes
  - The labels appear in applications. This avoids having to override them in the applications.
- Add a **description** and a **documentation**
  - Descriptions are for the model designers, documentations appear to end-users.
- Define **icons** for entities
  - They are used to graphically represent the entity in the application.

Using good names and labels for clearly defined business concepts always save you time later.

Using diagrams

- Use **diagrams** to edit ER Models
  - Use multiple diagrams to view subsets of complex models
  - You can expose diagrams to users in the application documentation.
- Diagram editor
  - **Toolbar** for display options
    - Show entities and attributes names, labels
    - Show references cardinality, names, labels
  - **Palette**
    - Create new entities/references
    - Add existing entities to the diagram



Semarchy

Demo

Entity creation

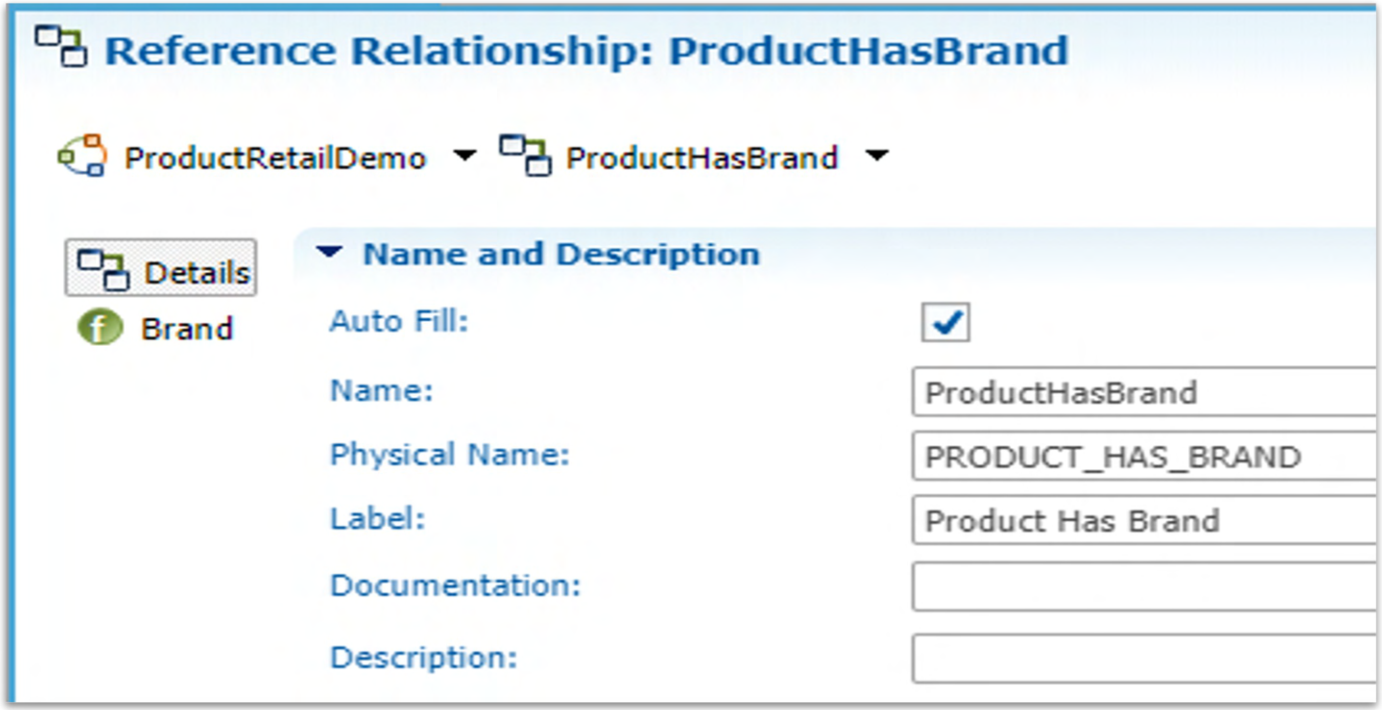
Semarchy

Modeling

Reference relationships

References Relate Two Entities

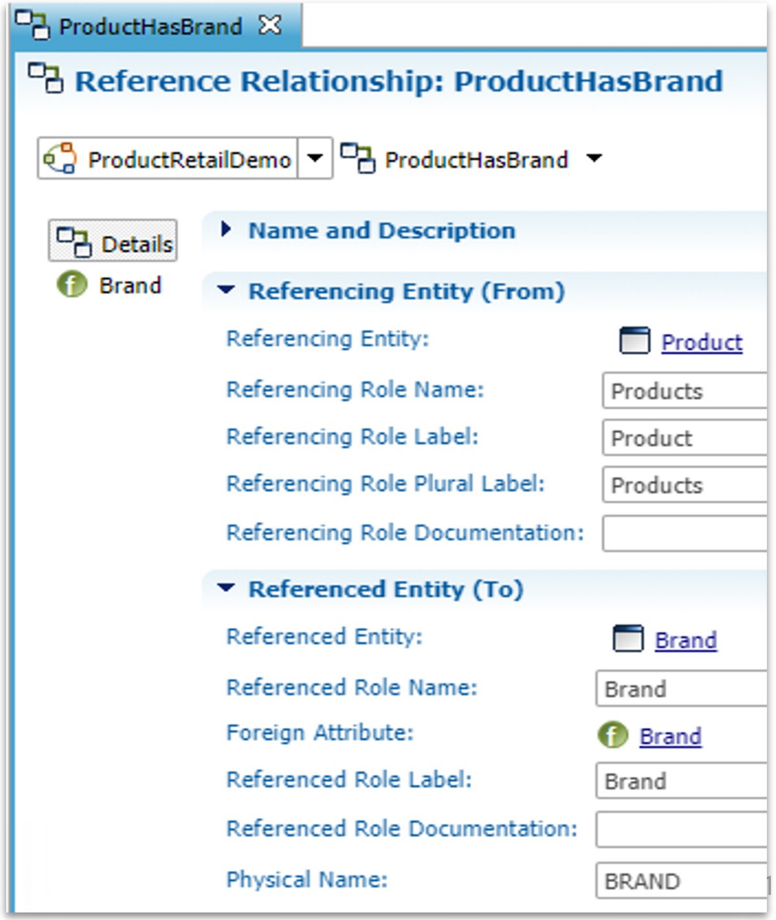
- Define a meaningful **Reference Name** and **Label**
  - e.g. *Product Has Brand*



Reference Relationships

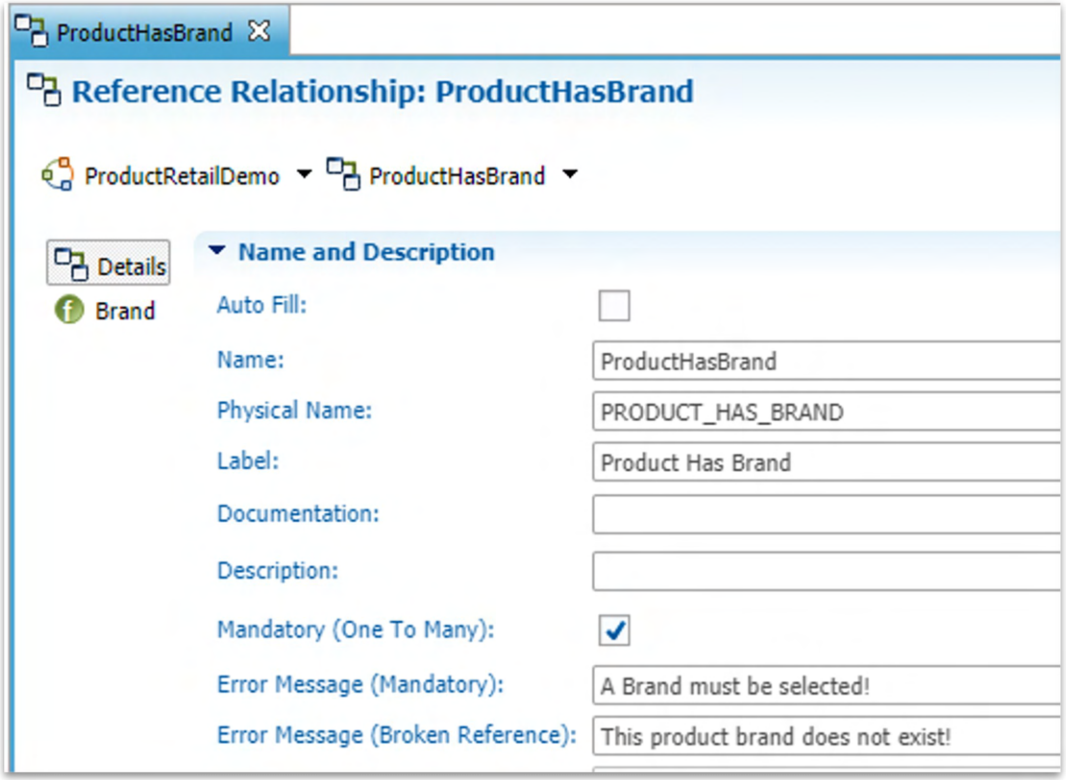
- Reference relates two entities who have different **role** in relationship
  - **Referencing** Entity is the child entity - *Product*
  - **Referenced** Entity is the parent entity - *Brand*

- Define meaningful **Role Names:**
  - **Referencing Role Name -** *Products*
  - **Referenced Role Name -** *Brand*

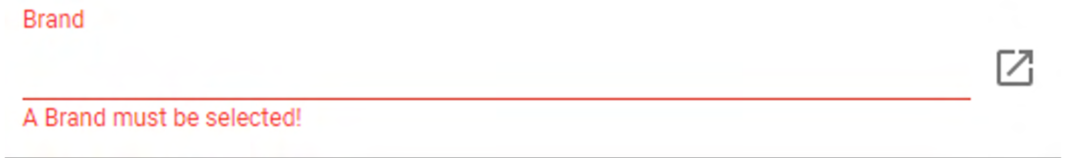


Reference Relationships

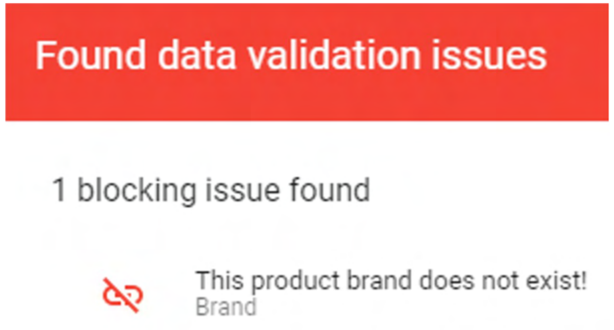
Define meaningful error messages



- For empty references:

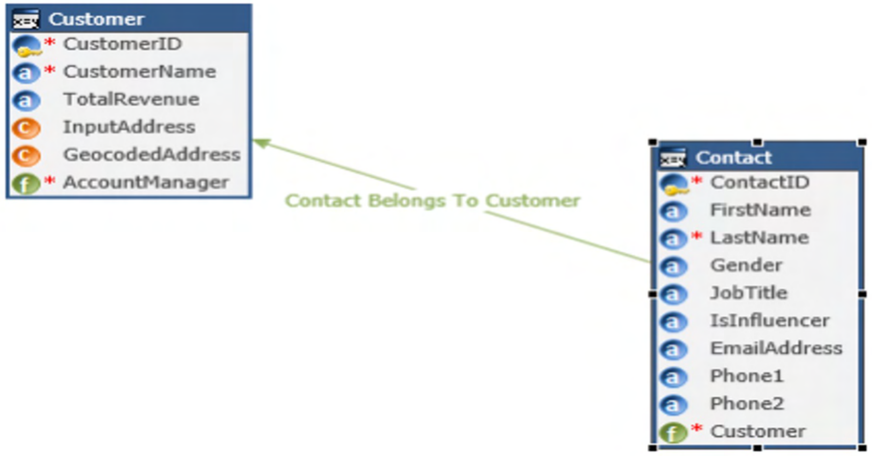


- For invalid references:



References - Common patterns

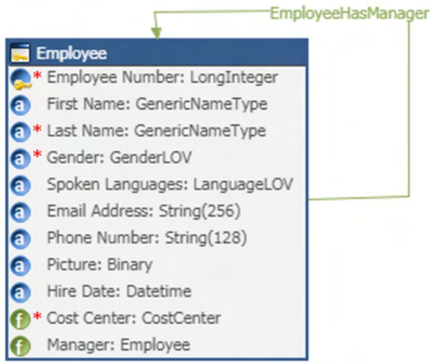
- 1-N parent-child



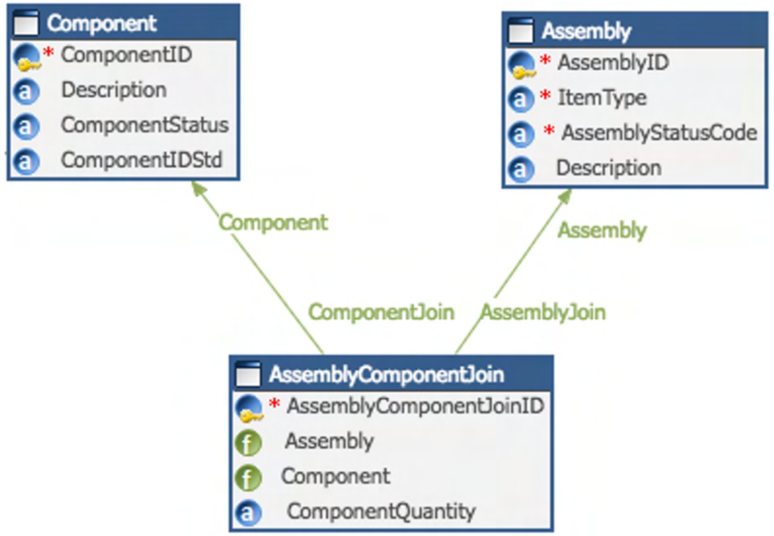
- Fixed-level hierarchy

References - Common patterns

- Ragged hierarchy (self-join)



- Join table (N-M relationship)



Demo

References

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Modeling

Record management

Record Management

Available actions and features for records depends on:

- The entity type (Basic, fuzzy or ID-matched)
- The reference relationships
- The rules applied to the entity and attributes
    - Enrichers, validations (E.g.: mandatory)
    - Matcher for fuzzy matched entities
    - Survivorship rules for ID and Fuzzy-matched entities
- The entity configuration
    - Historization
    - Delete

Historization

- Can be activated (optional) for each entity
  - Golden data can be historized for all entity types
  - Available for master record only with ID and fuzzy matched entities
- Allow users
  - to browse the history for a given record
  - browse the entire hub as of date.
- History is available in REST APIs and SQL APIs

Be careful with this feature as it may generate large volumes of data.

Browsing the history

Select Date / Stop Browsing History

Depends on application configuration

Non-historized entity

Record to History Navigation

Enables querying and compound views, such as “Current record with history”